

Isaac McDowell

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Education

Florida State University – BS in Computer Science

2021 - 2026

- **Relevant Coursework:** *Data Structures & Algorithms, Software Engineering, Database Systems, Object-Oriented Programming, Introduction to AI / Machine Learning, Web Development, Offensive Computer Security*

Projects

Smart Fitness App | *Flask, PostgreSQL, Python, Chart.js*

github.com/merleezy/smart-fitness-tracker

Solo – full-stack development

- Built and deployed a fitness app on Railway using Flask and PostgreSQL, enabling users to log meals by type, track workouts, and monitor weight over time.
- Reworked the recommendation engine into a live insight system that evaluates user meal, workout, and weight data across six rule-based patterns to generate personalized coaching insights.
- Integrated the USDA FoodData API for real-time food search with automatic macro extraction for 600,000+ items.
- Created an analytics dashboard with Chart.js to visualize weight trends and macronutrient breakdowns.

Tokyo Bloom - Restaurant Web App | *HTML/CSS, JavaScript, PHP, MySQL*

github.com/merleezy/tokyo-bloom

Team of 3 – full-stack development

- Designed and built the frontend UI, including an animated hero slider and Google Maps integration.
- Implemented double-booking prevention logic for the reservation system and fixed a UI bug where already-booked time slots were not visually reflected to users.
- Implemented security features including CSRF protection, input sanitization to prevent XSS attacks, and prepared statements to prevent SQL injection.
- Proposed and led an MVC restructure to improve codebase organization across our team.

Undead Alliance – Top-Down Zombie Shooter | *Unity, C#, .NET*

Team of 4 – UI & game development

- Designed UI elements and character animation systems in Unity to give players clearer feedback.
- Implemented inventory, leveling, and progression systems in C# and helped establish Git branching practices that reduced merge conflicts.

Experience & Involvement

Association for Computing Machinery (ACM), Member

2022 – 2025

- Participated in competitive programming competitions and workshops, practicing algorithm design and problem-solving under time constraints.

Cybersecurity Club at FSU, Member

2022 – 2024

- Competed in CTF events and attended workshops on web security, cryptography, and penetration testing.

Front Desk Staff, YMCA – Yulee, FL

2021

- Managed front desk operations, including member check-ins, phone and in-person inquiries, and coordination with staff in a fast-paced environment.

Skills

Languages: JavaScript, Python, C++, C#

Frontend: React, HTML/CSS, Chart.js

Backend: Node.js, Express, Flask, MongoDB, MySQL, SQLite

Tools: Git, Docker, Linux, Appwrite, Vite, Vercel